डॉ.श्यामाप्रसाद मुखर्जी अंतरराष्ट्रीय सूचना प्रौद्योगिकी संस्थान
DR.SHYAMA PRASAD MUKHERJEE INTERNATIONAL
INSTITUTE OF INFORMATION TECHNOLOGY, NAYA RAIPUR
(A Joint Initiative of NTPC & Govt. of Chhattisgarh)

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B.Tech - DSAI- UG curriculum

(With effect from July 2025)

Total Credit : 160

Lecture-L: 1 Credit = 1 hr/ week

Tutorial-T: 1 Credit = 1hr/week

Practical-P: 1 Credit = 2 hr/week

Scholarly Engagement Hours-S: 0 Credit= 2 hr/week

Curriculum

Semester 1 Total Credit : 21					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Linear Algebra and Matrix Analysis	MA-L-*xx*-C	BSC	3	2–1–0-0
2	Calculus	MA-L-*xx*-C	BSC	3	2–1–0-0
3	Problem Solving with C Programming	CS-L-*xx*-C	ESC/VSEC	3	2–0–2-2
4	Digital Electronics using Verilog	EC-L-*xx*-C	ESC	3	2–0–2-0
5	International Language Competency	HS-L-*xx*-C	HSMC (AEC)	2	1–0–2-2
6	IT Workshop : Full Stack Prototyping with AI Support	DA-L-*xx*-C	VSEC	2	1-0-2-2
7	Entrepreneurship and Design Thinking	HS-L-*xx*-C	HSMC	2	1-0-2-2
8	Internet of Things	EC-L-*xx*-C	VSEC	2	1-0-2-2
9	Environmental Studies	HS-L-*xx*-C	HSMC (VEC)	1	1-0-0-0
			Total	**21**	**14–2–10** **26**

Semester 2 Total Credit : 21					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Quantum Mechanics	PH-L-*xx*-C	BSC	3	2–1–0-0
2	Probability and Statistics	MA-L-*xx*-C	BSC	3	2-1-0-0
3	Data Structures	CS-L-*xx*-C	ESC/VSEC	3	2–0–2-2
4	Signals and Systems	EC-L-*xx*-C	ESC	3	2–0–2-0
5	Introduction to AIML	DA-L-*xx*-C	ESC/VSEC	2	1–0–2-2
6	Communication and Discourse Strategies	HS-L-*xx*-C	HSMC (AEC)	2	1–0–2-2
7	Human Values and Professional Ethics	HS-L-*xx*-C	HSMC (VEC)	2	2-0-0-2
8	3D Printing and Prototype Design	EC-L-*xx*-C	VSEC	2	1-0-2-2
9	Constitution of India	HS-L-*xx*-C	HSMC (VEC)	1	1-0-0-0
			Total	**21**	**13–2–12** **27**

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Exit: After successful completion of one year (first two semesters), a student may get an exit option as per ordinance. They need to do **Skill Based Courses of 6 credits additional related Web Development and Hardware Prototyping**, in summer, before exit. Department will provide a list of such courses.

Semester 3 Total Credit : 23					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Discrete Mathematics	MA-L-*xx*-C	BSC	3	2–1–0-0
2	Design and Analysis of Algorithms	CS-L-*xx*-C	PCC	3	2–0–2-2
3	Database Management Systems	CS-L-*xx*-C	PCC	3	2–0–2-2
4	Operating Systems	CS-L-*xx*-C	PCC	3	2–0–2-2
5	Object Oriented Programming	CS-L-*xx*-C	PCC	2	1-0-2-2
6	Optimization Methods	MA-L-*xx*-C	PSC	2	2-0-0-0
7	Communication Systems	EC-L-*xx*-C	PSC	2	1-0-2-0
8	*Multi-Disciplinary Minor-1*	MD-L-*xx*-E	MDM	3	3–0–0-0
9	Course on Indian Knowledge Systems	HS-L-*xx*-C	HSMC (AEC)	2	2–0–0-0
			Total	**23**	**17–1–10 28**

Semester 4 Total Credit : 22					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Graph Theory	MA-L-*xx*-C	PCC	3	2–0–2-2
2	AI Foundations	DA-L-*xx*-C	PCC	3	2–0–2-2
3	Applied Machine Learning	DA-L-*xx*-C	PCC	3	2–0–2-2
4	Pattern Recognition	DA-L-*xx*-C	PCC	3	2–0–2-0
5	Statistical Data Analysis	MA-L-*xx*-C	PCC	3	2–0–2-0
6	Data Visualization	DA-L-*xx*-C	PCC	2	1-0-2-0
7	Data Mining	MA-L-*xx*-C	PCC	2	1-0-2-0
7	*Multi-Disciplinary Minor-2*	MD-L-*xx*-E	MDM	3	3–0–0-0
8	*Experiential Learning through Mini Project*	DA-L-*xx*-C	EL	2	0-0-2-2
9	Community Services NSS / ISR Activities	HS-L-*xx*-C	HSMC (CEA)	2	0-0-0-2
			Total	**22**	**15–0–12 27**

Exit: After successful completion of 4 semesters, a student may get an exit option as per ordinance. They need to do **Skill Based Courses of 6 credits, additional**, in summer, before exit. Department will provide a list of such courses.

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Semester 5 Total Credit : 21					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Computer Vision	CS-L-*xx*-C	PCC	3	2–0–2-2
2	Natural Language Processing	DA-L-*xx*-C	PCC	3	2–0–2-2
3	Deep Learning	DA-L-*xx*-C	PCC	3	2–0–2-0
4	Robotics	DA-L-*xx*-C	PCC	3	2–0–2-2
5	Reinforcement Learning	DA-L-*xx*-C	PCC	3	2-0-2-2
5	Project - I (Research Methodology)	DA-L-*xx*-C	ELC	2	0–0–4-2
6	*Multi-Disciplinary Minor-3*	MD-L-*xx*-E	MDM	3	3–0–0-0
7	Introduction to Finance	HS-L-*xx*-C	HSMC (AEC)	1	1-0-2-0
			Total	**21**	**13–0–16 29**

Semester 6 Total Credit : 20					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Generative and Agentic AI Systems	DA-L-*xx*-C	PCC	3	2–0–2-2
2	Immersive Technologies: AR, VR and XR	DA-L-*xx*-C	PCC	2	1-0-2-0
2	Project - II	DA-L-*xx*-C	ELC	2	0–0-4-0
3	Open Elective - I	OE-L-*xx*-E	OE	3	3-0-0-0
4	Program Elective - I	PE-L-*xx*-E	PEC	3	3–0–0-0
5	Program Elective - II	PE-L-*xx*-E	PEC	3	3–0–0-0
	Program Elective - III	PE-L-*xx*-E	PEC	3	3-0-0-0
6	*Multi-Disciplinary Minor-4*	MD-L-*xx*-E	MDM	3	3–0–0-0
7	Indian/Foreign Language *Regional Language, Sanskrit, German, Japanese, French*	HS-L-*xx*-C	HSMC (AEC)	1	1-0-2-0
8	Co-Curricular & Extracurricular Activities	HS-L-*xx*-C	CCA	2	-
			Total	**25**	**15–0–8-0 23**

Exit: After successful completion of 6 semesters, a student may get an exit option as per ordinance after completion of the summer semester internship (3 credits) and additional 3 credit courses in summer.

Summer Semester Total Credit : 3					
Sl.No.	Course Name	Code	Type	Credit	
1	Internship		ELC	3	*Credit will be added in VII Sem.*

Note: Internship will be evaluated in the beginning of seventh semester. Its credit and grades will be reflected in the 7th semester Grade sheet.

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Semester 7 Total Credit : 21					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Minor Project	CS-L-*xx*-C	ELC	2	0–0-4-0
2	Program Elective – III	PE-L-*xx*-E	PEC	3	3–0–0-0
3	Program Elective - IV	PE-L-*xx*-E	PEC	3	3–0-0-0
4	Open Elective - II	OE-L-*xx*-E	OEC	3	3–0-0-0
5	*Multi-Disciplinary Minor-5*	MD-L-*xx*-E	MDM	3	3–0–0-0
6	*History of Indian Civilizations/ Kautilya's Arthashastra/ Vedic Mathematics/ Vedic Corpus/ Wisdom from the Ages/ Panini's Grammar*	HS-L-*xx*-C	HSMC (IKS)	2	2–0-0-0
7	Internship (Summer Semester)	CS-L-*xx*-C	ELC	3	0-0-0-6
			Total	**19**	**14–0–4** **18**

Semester 8 Total Credit : 12					
Sl.No.	Course Name	Code	Type	Credit	Hours L-T-P-S
1	Major Project	CS-L-*xx*-C	ELC	6	-
			Total	**6**	

* 8th Semester courses may be allowed to join via MOOC / NPTEL etc. Major projects may be completed as Internship cum projects.

Honors

- "Honors" is an additional credential a student will earn if he/ she opts for the extra 15 credits needed for this in his/ her own discipline. The concerned department specifies the course requirements for earning the Honors.
- BTech- DSAI- Honor Course: Minimum 8 CGPI and No backlog.
- After 3rd semester, students may choose extra credits for Honors
- The 8th Semester project must be an academic project. At least one research publication in Tier I/II conferences or in SCI Journals communicated or one patent filed.

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Multidisciplinary Minor (Tentative) Streams

1. Digital Forensics

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Introduction to Digital Forensics	3	1. Dr. Ruhul Amin 2. Dr. Satyanarayana Vollala
2	Computer and File System Forensics	3	
3	Network and Cloud Forensics	3	
4	Mobile and IoT Device Forensics	3	
5	Malware Analysis and Forensic Reporting	3	

2. Business Informatics

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Fundamentals of Business Informatics	3	1. Dr. Amit Agarwal 2. Dr. Vinay Kumar
2	Enterprise Resource Planning (ERP) Systems	3	
3	Business Data Analytics	3	
4	Business Process Management	3	
5	Digital Transformation and Strategic IT Management	3	

3. Digital Humanities

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Introduction to Digital Humanities	3	1. Dr. Shresha Yadav 2. Dr. Shrivishal Tripati
2	Digital Text Analysis and Corpus Linguistics	3	
3	Digital Archives, Mapping, and Storytelling	3	
4	Data, Culture, and Algorithmic Interpretation	3	
5	Research Methods and Project in Digital Humanities	3	

4. Electrical Vehicle Technology

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Fundamentals of Electric Vehicles	3	1. Dr. Bipin Mandi 2. Dr. Lakender Murmu
2	Electric Drive Systems and Motor Control	3	
3	Battery Management Systems (BMS) and Charging Infrastructure	3	
4	Power Electronics for EVs	3	
5	EV System Integration and Autonomous Features	3	

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5. Quantum Technologies

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Introduction to Quantum Computing	3	1. Prof. Rajarshi Mahapatra 2. Dr. P P Paltani
2	Quantum Materials and Devices	3	
3	Fundamentals of Photonics	3	
4	Quantum Information Theory	3	
5	Quantum Communication and Networks	3	

6. Innovation, Entrepreneurial and Venture Development (IEV)

Sl. No.	Name of the Course	Credits	Faculty Coordinators:
1	Creativity, Innovation, and Entrepreneurship Fundamentals	3	1. Prof. Srinivasa K G 2. Dr. Amit Agarwal
2	Business Model Development and Validation	3	
3	Technology Prototyping and Product Development	3	
4	Startup Finance, Legal, and Regulatory Framework	3	
5	Venture Launch and Growth Strategies	3	

Note: Department will float other minors in future.